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Project Title	Mathematical Customer Lifetime Value (CLV) Projection via BG/NBD Model from Scratch
Week	Week 1

WEEK-1 Technical Report

Task Name

Mathematical Customer Lifetime Value (CLV) Projection via BG/NBD Model from Scratch

Abstract

Customer Lifetime Value (CLV) is a key metric used by businesses to estimate the future value generated by customers. This project presents a complete implementation of the **Beta Geometric/Negative Binomial Distribution (BG/NBD)** model from first principles using **NumPy** and **SciPy** without relying on specialized marketing analytics libraries. Model parameters are estimated through **Maximum Likelihood Estimation (MLE)** using the **L-BFGS-B optimization algorithm**, enabling efficient prediction of future customer transactions and lifetime value.

Introduction

Understanding customer purchasing behavior is essential for customer relationship management, marketing optimization, and revenue forecasting. The BG/NBD model is one of the most widely used probabilistic models for predicting repeat purchase behavior.

Instead of using existing marketing packages, this project develops every mathematical component manually, including the likelihood function, parameter estimation, and prediction

routines. The implementation demonstrates how probabilistic modeling and numerical optimization can be combined to estimate customer lifetime value from transaction histories.

Theory

The BG/NBD model is based on the following probabilistic assumptions:

- Customer purchases follow a **Poisson process**.
- Individual transaction rates vary according to a **Gamma distribution**.
- After each purchase, customers may become inactive with probability **p**.
- Dropout probabilities follow a **Beta distribution**.
- Customer purchasing behavior is heterogeneous across the population.

The four parameters estimated are:

- **r** - Shape parameter of transaction rate distribution
- **α (alpha)** - Scale parameter of transaction rate distribution
- **a** - Shape parameter of dropout probability
- **b** - Shape parameter of dropout probability

These parameters collectively describe purchasing frequency and customer retention.

Objectives

The objectives of this project are:

- Implement the BG/NBD mathematical model entirely from scratch.
 - Construct the vectorized negative log-likelihood function.
 - Estimate the parameters **r**, **α** , **a**, and **b** using Maximum Likelihood Estimation.
 - Perform optimization using SciPy's **L-BFGS-B** algorithm.
 - Predict expected future customer transactions.
 - Estimate Customer Lifetime Value (CLV).
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Methodology

The implementation consists of the following stages:

Step 1: Transaction Data Preparation

Customer transaction histories are summarized into:

- x - Number of repeat purchases
- t_x - Time of last purchase
- T - Customer observation period

These summaries provide sufficient statistics required by the BG/NBD model.

Step 2: Mathematical Likelihood Construction

A vectorized negative log-likelihood function is developed using NumPy. The implementation avoids Python loops to maximize computational efficiency.

Step 3: Parameter Estimation

The four unknown parameters

- r
- α
- a
- b

are estimated by minimizing the negative log-likelihood using **SciPy's L-BFGS-B optimizer**, which supports bound constraints and converges efficiently for nonlinear optimization problems.

Step 4: Prediction

Using the optimized parameters, the model calculates:

- Probability that a customer is still active
- Expected number of future purchases
- Projected Customer Lifetime Value (CLV)

Dataset

A synthetic customer transaction dataset is generated to validate the implementation and verify parameter estimation.

Each customer record is summarized into:

- Purchase frequency (x)
- Recency (t_x)
- Observation period (T)

The same workflow can be directly applied to real-world e-commerce or retail transaction datasets after converting purchase histories into BG/NBD summary statistics.

Mathematical Likelihood

The BG/NBD model estimates parameters by maximizing the likelihood of observing customer transaction histories.

Maximum Likelihood Estimation (MLE) is performed by minimizing the negative log-likelihood:

$$\theta = (r, \alpha, a, b)$$

where

- r and α describe transaction rate heterogeneity,
- a and b describe customer dropout behavior.

SciPy's **L-BFGS-B** algorithm iteratively updates these parameters until convergence.

Complexity Analysis

The implementation is fully vectorized using NumPy.

Performance characteristics include:

- **Likelihood evaluation:** $O(N)$
- **Memory complexity:** $O(N)$
- **Optimization:** Multiple $O(N)$ likelihood evaluations during iterative convergence

The absence of explicit Python loops significantly improves scalability, allowing the algorithm to process millions of customer transaction summaries efficiently.

Results

The implementation successfully estimates the BG/NBD model parameters:

- r
- α

- a
- b

Using these estimates, the model computes:

- Customer probability alive
- Expected future transaction counts
- Customer Lifetime Value (CLV)

The vectorized implementation demonstrates fast convergence and numerical stability while maintaining computational efficiency.

Applications

The BG/NBD model can be applied in:

- Customer Lifetime Value Prediction
 - Customer Retention Analysis
 - Marketing Campaign Optimization
 - Customer Segmentation
 - Subscription Services
 - E-commerce Recommendation Systems
 - CRM Analytics
 - Revenue Forecasting
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Conclusion

This project demonstrates a complete mathematical implementation of the BG/NBD model without relying on specialized marketing libraries. By combining vectorized NumPy computations with SciPy's **L-BFGS-B** optimization algorithm, the implementation efficiently estimates transaction behavior, customer retention, and future purchasing activity. The developed solution is computationally scalable, numerically stable, and suitable for educational, research, and real-world customer analytics applications.